Chapter 1

Computer Abstractions and Technology

Understanding Performance

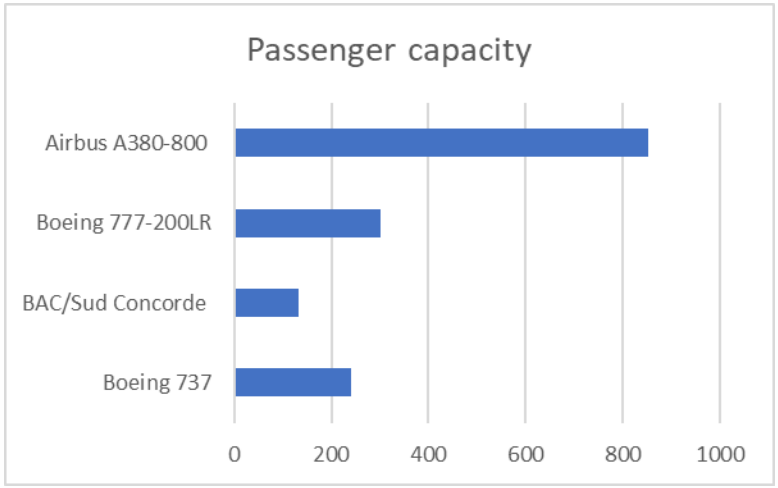
- Algorithm
  - Determines number of operations executed
- Programming language, compiler, architecture
  - Determine number of machine instructions executed per operation
- Processor and memory system
  - Determine how fast instructions are executed
- I/O system (including OS)

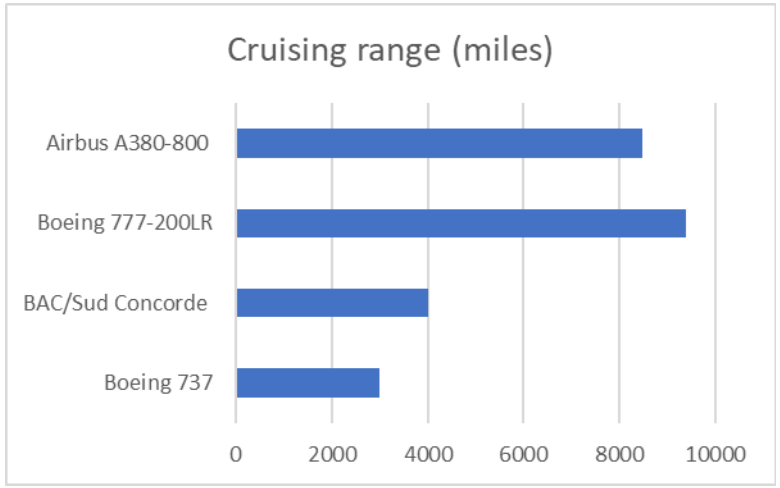
Q1. Can the same source code have different performance on the same computer? Why?

A1. Yes. Different compilers or optimization settings may generate different machine instructions.

Defining Performance

- Which airplane has the best performance?





Q2. Which server has better performance?

A: 1 request takes 1 sec., up to 100 requests/min

B: 1 request takes 2 sec., up to 300 requests/min

A2. It depends on the performance metric.

Response time: Server A is better.

Throughput: Server B is better.

Response Time and Throughput

- Response time
  - How long it takes to do a task
- Throughput
  - Total work done per unit time
    - e.g., tasks/transactions/… per hour
- How are response time and throughput affected

Q3. Which metric matters more for each task: response time or throughput?

A. Online gaming

B. B. Processing millions of bank transactions

A3.

A. Response time

B. Throughput

Relative Performance

- Define Performance = 1/Execution Time
- "X is n time faster than Y"

$$\text{Performance}_X / \text{Performance}_Y$$
$$= \text{Execution time}_Y / \text{Execution time}_X = n$$

- Example: time taken to run a program
  - 10s on A, 15s on B
  - Execution Time_B / Execution Time_A

Q4. Computer A has 50% less execution time than Computer B. How much faster is A?

A4. A is 2x as fast, or 2 times faster, or 100% faster.

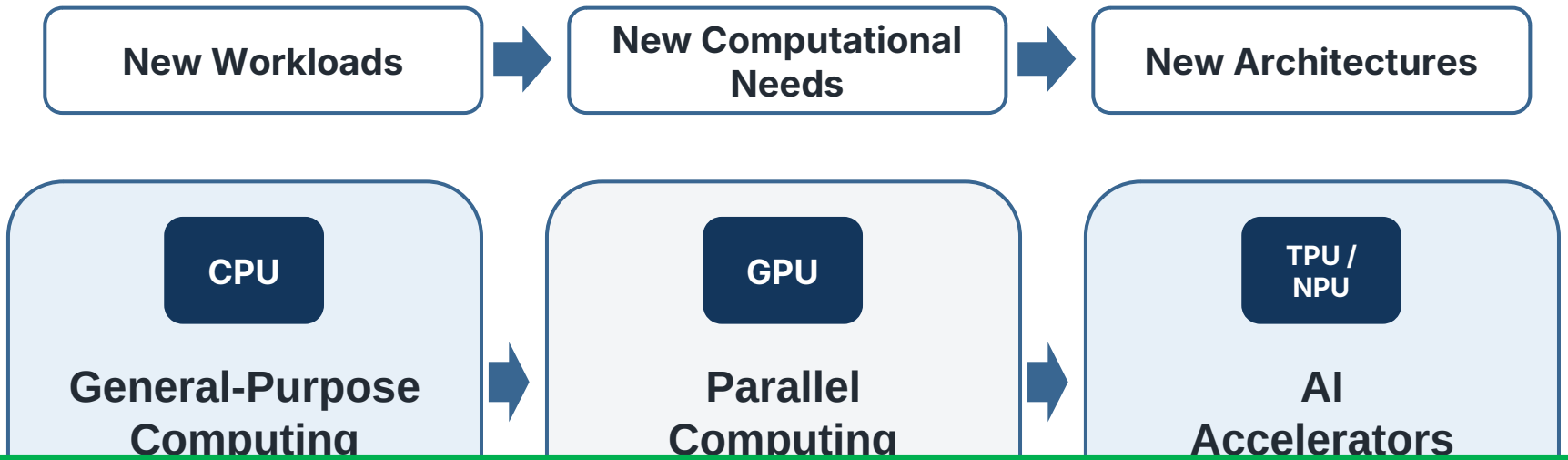
Measuring Execution Time

- Elapsed time
  - Total response time, including all aspects
    - Processing, I/O, OS overhead, idle time
  - Determines system performance
- CPU time
  - Time spent processing a given job

Q5. A program has an elapsed time of 20 s but uses only 5 s of CPU time. Would a 2× faster CPU reduce the elapsed time to 10 sec?

A5. No. Elapsed time should be 5/2+(20−5)=17.5 sec

The Rise of AI Computing



Q6. Venture a guess: What types of tasks are CPUs, GPUs, and AI accelerators best suited for?

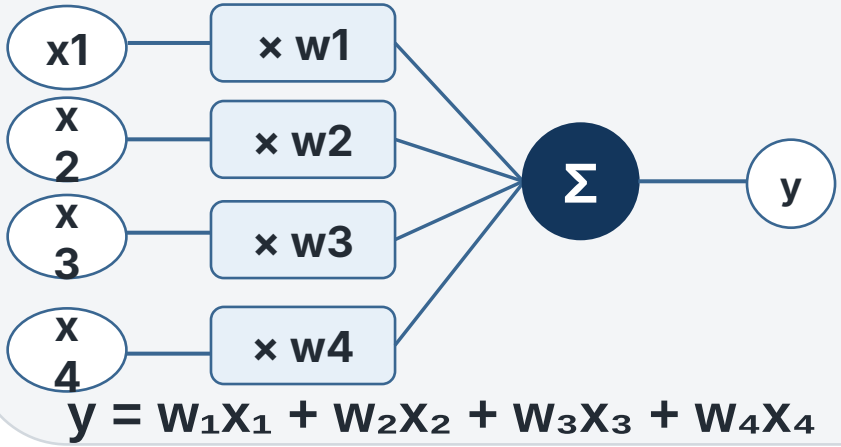
A6.

CPU: General-purpose and control-intensive tasks

GPU: Highly parallel tasks

AI Accelerator: Specialized AI tasks

Why Does AI Need Specialized Hardware?



Many multiplications and additions

- Similar repeated operations
- Many operations can run at the same time

Q7. What real-world tasks can benefit from both parallelism and specialization?

A7.

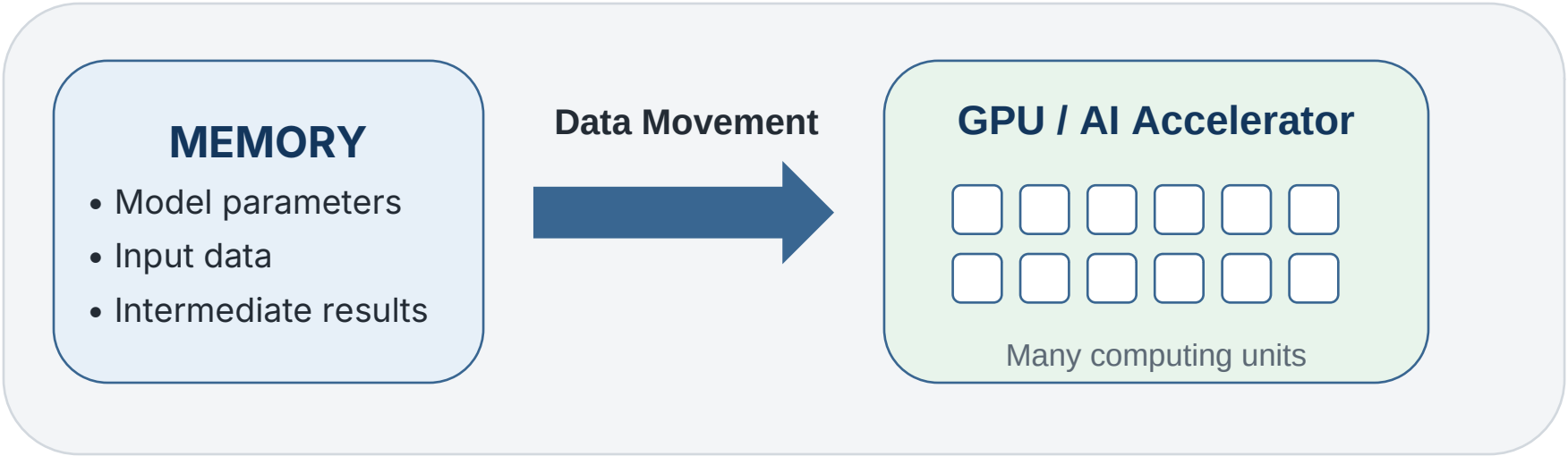
Image recognition: many pixels and features

LLM inference: massive matrix operations

Autonomous driving: multiple cameras and sensors

Computing Is Not the Only Bottleneck

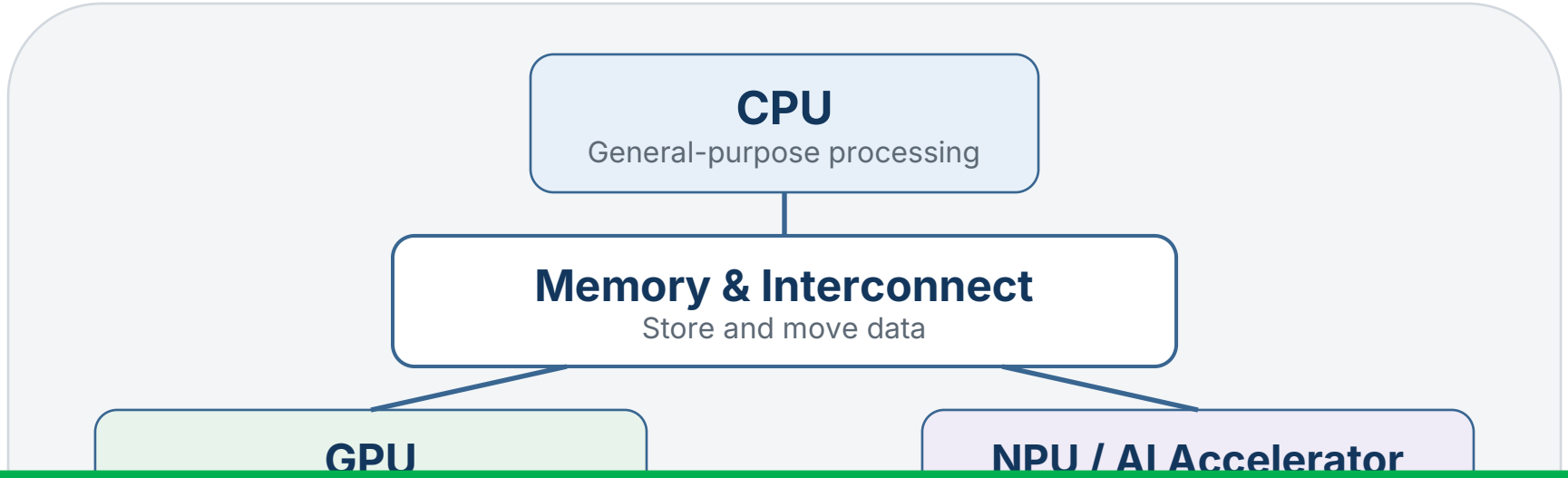
More computing units help only when the required data can arrive fast enough.



Q8. Which system finishes the task faster? Both systems process 100 GB of data.
A: Data transfer = 10 GB/s, Compute time = 2s
B: Data transfer = 50 GB/s, Compute time = 6 s
A8. System B is faster: 8 sec vs. 12 sec. A faster GPU does not guarantee better overall performance if data transfer is slow.

Modern Computer Architecture

What determines the performance of a modern computing system?



Q9. What has changed from traditional to modern computer architecture?

A9.

Traditional view: CPU, Memory, Input, Output

Modern view: adds/emphasizes GPUs, AI accelerators, and high-performance interconnects.

Computing is increasingly parallel and specialized

Section 1-2 Quick Questions

Q1. Can the same source code have different performance on the same computer? Why?

Q2. Which server has better performance?

Q3. Which metric matters more for each task: response time or throughput?

Q4. Computer A has 50% less execution time than Computer B. How much faster is A?

Q5. A program has an elapsed time of 20 s but uses only 5 s of CPU time. Would a 2× faster CPU reduce the elapsed time to 10 sec?

Section 1-2 Quick Questions (cont.)

Q6. What types of tasks are CPUs, GPUs, and AI accelerators best suited for?

Q7. What real-world tasks can benefit from both parallelism and specialization?

Q8. Which system finishes the task faster? Both systems process 100 GB of data.

Q9. What has changed from traditional to modern computer architecture?

Exercise 1 | Choosing Hardware for Different Workloads

Match each workload with the most suitable processor type. Explain your choice briefly.

2. Rendering millions of pixels for a 3D scene
3. Neural-network inference on an edge device
1. Operating-system control and general applications

Answer

1. CPU — flexible, general-purpose, and good for control-intensive tasks.
2. GPU — well suited to highly parallel computation.
3. AI accelerator— specialized for neural-network operations.

Key Point: Different workloads benefit from different processor architectures.

Exercise 2 | Response Time vs. Throughput

A web service is upgraded. Before the upgrade, each request takes 4 s and the server completes 120 requests/min. After the upgrade, each request still takes 4 s, but the server completes 240 requests/min.

Which performance metric improved? By how much?

Answer

Response time: unchanged at 4 s/request.
Throughput: 120 → 240 requests/min = 2× throughput (100% increase).
Therefore, the upgrade improves throughput, not response time.

Key Point: Higher throughput does not necessarily mean a shorter response time.

Exercise 3 | Relative Performance

Computer A is 50% faster than Computer B. Computer B executes a program in 18 s.

1. How many times faster is A than B?
2. What is A's execution time?
3. Is A's execution time 50% shorter than B's?

Answer

1. 50% faster = 1.5× performance.
2. T_A = 18 / 1.5 = 12 s.
3. No. The reduction is (18 − 12) / 18 = 33.3%.

Key Point: A percentage increase in performance is not the same percentage decrease in execution time.

Exercise 4 | Elapsed Time vs. CPU Time

A program has an elapsed time of 30 s, including 12 s of CPU time. The CPU is replaced by one that executes the CPU portion 3× as fast. Assume all non-CPU time is unchanged.

What is the new elapsed time?

Answer

Non-CPU time = 30 − 12 = 18 s
New CPU time = 12 / 3 = 4 s
New elapsed time = 4 + 18 = 22 s

Key Point: Speeding up the CPU affects only the CPU portion of elapsed time.

Exercise 5 | Integrated Performance Analysis

Question

A web server is upgraded from System A to System B.

Metric	System A	System B
Elapsed time / request	12 s	8 s
CPU time / request	8 s	5 s
Throughput	100 requests/min	150 requests/min

1. Which system has better response time? By what factor?
2. By what percentage does throughput improve?
3. Based on CPU time, how many times as fast is System B as System A?
4. Does CPU improvement alone explain the elapsed-time improvement? Explain.

Exercise 5 | Integrated Performance Analysis — Answer

Answer

1. Response time: System B is better. 12 / 8 = **1.5×** the response-time performance of A.

System A: 12 − 8 = 4 s | System B: 8 − 5 = 3 s

Both CPU time and non-CPU time are reduced.

2. Throughput: (150 − 100) / 100 × 100% = 50% improvement.

3. CPU performance: P_B / P_A = T_A / T_B = 8 / 5 = 1.6×.

4. No. Non-CPU time also improves:

Key Point: Response time, throughput, relative performance, and CPU time describe different aspects of system performance.